D3. Zero-Shot and In-Context Learning Design

ICL implementation, comparison protocol, and evidence boundary.

Field	Value
Project	Development of AI-Based Cyber Security Audit Assessment Against Benchmark
Scope	Objective 3; secure local/air-gapped RAG proof of concept
Report date	28 August 2026
Evidence rule	Implemented, tested and untested claims are labelled separately.

ICL experiment and design decision

Configuration	Examples supplied	Accuracy	Macro-F1	Decision
Zero shot	No demonstrations	75.00%	0.4697	Best Qwen2.5 control result
Retrieval few shot	4 semantically selected local examples	62.50%	0.4190	Works; not selected on this control
Fixed few shot	Fixed local examples	50.00%	0.3789	Not selected

Why examples remain a controlled variable

Few-shot examples are not report evidence. They are local formatting and reasoning demonstrations, selected separately from the retrieved audit report. Keeping that boundary explicit prevents an ICL example from silently introducing facts into an assessment answer. [E8][E10]

The observed zero-shot result means the present four-example bank did not improve this small Qwen2.5 evaluation. The final Qwen3 configuration retains retrieval few shot only as the existing production-shaped setting; zero shot should be retested with Qwen3 before a cross-model ICL conclusion is made. [E10]

Implemented strategies

Strategy	Implementation	Use
zero_shot	Direct assessment prompt with retrieved report evidence	Baseline without demonstrations
fixed_few_shot	Fixed examples from the local example bank	Controls example wording/count
retrieval_few_shot	Semantically selected examples from icl_example_bank.json	Current default; k=4 [E4][E8]

Design rationale

Examples show the required evidence-first decision pattern: answer Yes, No or Insufficient Evidence; cite retrieved chunks; provide a short justification. The example bank is local and is never treated as audit evidence for the report under assessment. [E8]

Retrieval-selected examples are a sensible implementation choice because audit questions vary by control, finding and wording. Their use is not yet empirically proven superior to zero-shot or fixed examples. [E4][E8]

Controlled comparison plan

Variable	Held constant	Changed	Measures
ICL strategy	Reports, labels, retrieval context, model, prompt, seed, temperature	zero-shot / fixed / retrieval	Accuracy, macro-F1, grounding, abstention safety, latency

Variable	Held constant	Changed	Measures
Example count	Same selection policy and examples where applicable	k = 0, 1, 2, 4, 6	Quality change and context cost
Example composition	Question set and evaluation harness	Positive, negative, missing-evidence, difficult examples	Per-class F1 and unsafe-answer rate

What was tested

A controlled Qwen2.5-7B comparison completed on the same 16 report-question pairs: zero shot achieved 75.00% accuracy and 0.4697 macro-F1; retrieval few shot k=4 achieved 62.50% and 0.4190; fixed few shot achieved 50.00% and 0.3789. [E10]

The 28 August verification also passed the offline mock end-to-end path and focused parser, routing/citation, and safety-gate tests. [E7]

Conclusion

Select zero shot as the strongest tested Qwen2.5 ICL condition. It must still be reconfirmed with Qwen3 before it is called the final Qwen3 ICL setting. [E10]

Worked in-context example pattern

Question	Evidence	Required verdict	Why this matters
Are backups tested quarterly?	Backups complete nightly; restore testing is not mentioned.	Insufficient Evidence	Demonstrates that silence is not a Yes or No claim.
Is segmentation implemented?	Payment subnet is reachable from corporate VLAN without an ACL.	No	Demonstrates a contradiction-backed negative answer.
Is password rotation enforced?	Privileged accounts rotate every 60 days.	Yes	Demonstrates threshold-based evidence use.

Review conclusion

The example bank defines output behaviour; it must never be mistaken for evidence about the report being assessed. The worked patterns cover affirmative, negative and missing-evidence cases so the model sees the required answer format without being given facts from the target audit report. [E8]

On the frozen Qwen2.5 control, adding the present examples did not improve accuracy: zero shot 75.00%, retrieval few shot 62.50%, fixed few shot 50.00%. The defensible decision is to retain zero shot as the tested control winner and label Qwen3 ICL selection as a required follow-up, not as an assumed transfer. [E10]

Implementation and retest requirement

Three ICL strategies are implemented in icl.py. Zero shot returns no demonstrations; fixed few shot chooses a class-diverse local set; retrieval few shot embeds the local question bank and selects the nearest four examples for the incoming benchmark question. The prompt builder inserts demonstrations before the report evidence and clearly separates the two. [E8]

That separation matters operationally. An example may teach the model how to answer a segmentation question, but it cannot prove that the target report has segmentation. The final verdict must always cite chunks retrieved from the target report, and the citation validator rejects IDs outside that returned context. [E7][E8]

The next controlled run should repeat zero-shot and retrieval-few-shot settings with Qwen3-8B, hold retrieval/prompt/seed fixed, and add independent No and Insufficient Evidence cases. Only then can ICL be selected for the final Qwen3 configuration rather than inherited from the current implementation. [E10]

Repository evidence cited

- [E7] Verification executed 28 August 2026: python -m unittest test_json_parser.py test_adaptive_rag.py test_security_gates.py (6 tests passed) and python test_harness_mock.py (offline mock end-to-end passed).
- [E8] src/{docling_parser.py,chunking.py,embed_index.py,pipeline.py,evaluation_harness.py,adaptive_rag.py,prompts.py,icl.py}: implementation inspection.
- [E10] results/final_validation/VALIDATION_SUMMARY.md and its cited run directories: controlled local LLM, ICL, prompt, retrieval, CRAG, chunking and chunk-parameter comparisons completed 28 August 2026.